

Chapter 11

Special Functions

LEGENDRE FUNCTIONS

The Legendre equation

$$(1 - x^2)P''(x) - 2xP'(x) + \lambda P(x) = 0,$$

It has regular singular points at $x = \pm 1$ and $x = \infty$, and therefore has a series solution about $x = 0$ that has a unit radius of convergence, i.e., the series solution will (for all values of the parameter λ) converge for $|x| < 1$. However, if λ has the value $l(l+1)$, with l an integer, the series become truncated after x^l , leaving a polynomial of degree l .

The generating function for the polynomial solutions of the Legendre ODE is given by

$$g(x, t) = \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n$$

To identify the scale that is given to P_n by Eq. (1), we simply set $x = 1$ in that equation, bringing its left-hand side to the form

$$g(1, t) = \frac{1}{\sqrt{1 - 2t + t^2}} = \frac{1}{1 - t} = \sum_{n=0}^{\infty} t^n \dots\dots\dots (2)$$

where the last step in Eq. (2) was to expand $1/(1 - t)$ using the binomial theorem. Comparing with Eq.(1), we see that the scaling it predicts is $P_n(1) = 1$.

Now, if we replace x by $-x$ and t by $-t$. The value of $g(x, t)$ in Eq. (1) is unaffected by this substitution, but the right-hand side takes a different form:

$$\sum_{n=0}^{\infty} P_n(x)t^n = g(x, t) = g(-x, -t) = \sum_{n=0}^{\infty} P_n(-x)(-t)^n, \dots\dots\dots(3)$$

showing that

$$P_n(-x) = (-1)^n P_n(x) \dots\dots\dots(4)$$

From this result it is obvious that $P_n(-1) = (-1)^n$, and that $P_n(x)$ will have the same parity as x^n .

Another useful special value is $P_n(0)$. Writing P_{2n} and P_{2n+1} to distinguish even and odd index values, we note first that because P_{2n+1} is odd under parity, i.e., $x \rightarrow -x$, we must have $P_{2n+1}(0) = 0$.

To obtain $P_{2n}(0)$

$$P_{2n}(0) = (-1)^n \frac{(2n-1)!!}{(2n)!!} \dots\dots\dots(5)$$

These results are important, so we summarize:

$P_n(x)$ has sign and scaling such that $P_n(1) = 1$ and $P_n(-1) = (-1)^n$.

$P_{2n}(x)$ is an even function of x ; $P_{2n+1}(x)$ is odd. $P_{2n+1}(0) = 0$, and $P_{2n}(0)$ is given by Eq. (5).

$P_n(x)$ is a polynomial of degree n in x , with the coefficient of x^n ; $P_n(x)$ contains alternate powers of x : $x^n, x^{n-2}, \dots, (x^0 \text{ or } x^1)$. From the fact that P_n is of degree n with alternate powers, it is clear that $P_0(x) = \text{constant}$ and that $P_1(x) = (\text{constant})x$. From the scaling requirements these must reduce to $P_0(x) = 1$ and $P_1(x) = x$.

Recurrence Formula

From the generating function equation we can generate recurrence formulas by differentiating $g(x, t)$ with respect to x or t .

$$\frac{\partial g(x, t)}{\partial t} = \frac{x - t}{(1 - 2xt + t^2)^{3/2}} = \sum_{n=0}^{\infty} n P_n(x) t^{n-1}$$

which we rearrange to

$$(1 - 2xt + t^2) \sum_{n=0}^{\infty} n P_n(x) t^{n-1} + (t - x) \sum_{n=0}^{\infty} P_n(x) t^n = 0,$$

and then expand, reaching

$$\begin{aligned} \sum_{n=0}^{\infty} nP_n(x)t^{n-1} - 2 \sum_{n=0}^{\infty} nxP_n(x)t^n + \sum_{n=0}^{\infty} nP_n(x)t^{n+1} \\ + \sum_{n=0}^{\infty} P_n(x)t^{n+1} - \sum_{n=0}^{\infty} xP_n(x)t^n = 0 \end{aligned}$$

Collecting the coefficients of t^n from the various terms and setting the result to zero, Then

$$(2n+1)xP_n(x) = (n+1)P_{n+1}(x) + nP_{n-1}(x), \quad n = 1, 2, 3, \dots$$

Above Equation permits us to generate successive P_n from the starting values P_0 and P_1 that we have previously identified.

For example,

$$2P_2(x) = 3xP_1(x) - P_0(x) \quad \longrightarrow \quad P_2(x) = \frac{1}{2}(3x^2 - 1)$$

By continuing this process

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= \frac{1}{2}(3x^2 - 1) \\ P_3(x) &= \frac{1}{2}(5x^3 - 3x) \\ P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3) \\ P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x) \\ P_6(x) &= \frac{1}{16}(231x^6 - 315x^4 + 105x^2 - 5) \\ P_7(x) &= \frac{1}{16}(429x^7 - 693x^5 + 315x^3 - 35x) \\ P_8(x) &= \frac{1}{128}(6435x^8 - 12012x^6 + 6930x^4 - 1260x^2 + 35) \end{aligned}$$

As before, the coefficient of each power of t is set to zero and we obtain

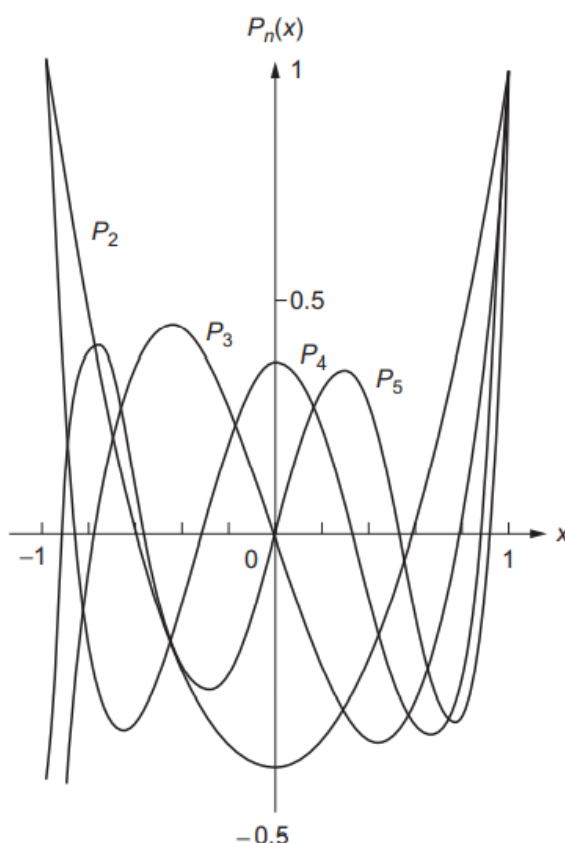
$$P'_{n+1}(x) + P'_{n-1}(x) = 2xP'_n(x) + P_n(x)$$

Numerous additional relations can be developed, including

$$\begin{aligned} P'_{n+1}(x) - P'_{n-1}(x) &= (2n+1)P_n(x), \\ P'_{n+1}(x) &= (n+1)P_n(x) + xP'_n(x), \\ P'_{n-1}(x) &= -nP_n(x) + xP'_n(x), \\ (1-x^2)P'_n(x) &= nP_{n-1}(x) - nxP_n(x), \\ (1-x^2)P'_n(x) &= (n+1)xP_n(x) - (n+1)P_{n+1}(x) \end{aligned}$$

The Legendre polynomial $P_n(x)$ has a global maximum on the interval $(-1, +1)$ at $x = 1$, with value $P_n(1) = 1$, and if n is even, also at $x = -1$. If n is odd, $x = -1$ will be a global minimum on this interval with $P_n(-1) = -1$. The maxima and minima of the Legendre

polynomials can be seen from the graphs:



Legendre – Polynomials

Example: The value of $P'_n(1)$ is given by:

- (a) $\frac{1}{2}n(n-1)$
- (b) $\frac{1}{2}n(n+1)$
- (c) $n(n-1)$
- (d) $n(n+1)$

solution: $(1-x^2)P''_n(x) - 2xP'_n(x) + n(n+1)P_n(x) = 0$

Put $x = 1$

$$\Rightarrow -2P'_n(1) + n(n+1)P_n(1) = 0$$

$$P'_n(1) = \frac{n(n+1)}{2}P_n(1), P_n(1) = 1$$

$$P'_n(1) = \frac{n(n+1)}{2}$$

Example: The value of $P''_n(1)$ is given by:

Solution: $(1-x^2)P''_n(x) - 2xP'_n(x) + (n+1)P_n(x) = 0$

$$\left((1-x^2)P'''_n(x) + P''_n(x)(-2x) - 2(xP''_n(x) + P'_n(x)) \right) + n(n+1)P'_n(x) = 0$$

Put $x = 1$

$$P''_n(1)[-2] - 2[P''_n(1) + P'_n(1)] + n(n+1)P'_n(1) = 0$$

$$\Rightarrow -2P''_n(1) - 2P''_n(1) - 2P'_n(1) + n(n+1)P'_n(1) = 0$$

$$-4P''_n(1) + (n(n+1) - 2)P'_n(1) = 0$$

$$4P''_n(1) = [n(n+1) - 2]P'_n(1) = \frac{(n(n+1)-2)}{2}n(n+1)$$

$$P''_n(1) = \frac{[n(n+1)-2]n(n+1)}{8}$$

Example: $f(x) = ax^2 + bx + c$, write it in the form of legendre polynomials

Solution: $ax^2 + bx + c = a_0P_0(x) + a_1P_1(x) + a_2P_2(x) + a_3P_3(x) + a_4P_4(x)$

$$= a_0 + a_1x + a_2\left(\frac{1}{2}(3x^2 - 1)\right) + a_3 \cdot \frac{1}{2}(5x^3 - 3x) + a_4 \frac{1}{8}(35x^4 - 30x^2 + 3)$$

$$p_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1)$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x)$$

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)$$

$$\frac{2P_2 + P_0}{3} = x^2$$

So $ax^2 + bx + c = a\left[\frac{2P_2(x) + P_0(x)}{3}\right] + bP_1(x) + P_0(x)$

Rodrigues Formula

Orthogonal polynomials could be described by Rodrigues formulas:

$$P_n(x) = \frac{1}{2^n n!} \left(\frac{d}{dx} \right)^n (x^2 - 1)^n$$

Orthogonality

Because the Legendre ODE is self-adjoint and the coefficient of $P''(x)$, namely $(1 - x^2)$, vanishes at $x = \pm 1$, its solutions of different n will automatically be orthogonal with unit weight on the interval $(-1, 1)$,

$$\int_{-1}^1 P_n(x) P_m(x) dx = 0, \quad (n \neq m)$$

Because the P_n are real, no complex conjugation needs to be indicated in the orthogonality integral.

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n + 1}$$

And

$$\int_{-1}^1 P_n(x) P_m(x) dx = \frac{2\delta_{nm}}{2n + 1}.$$

This result can also be obtained using the Rodrigues formulas for P_n and P_m .

Example:

$$\begin{aligned} \int_{-1}^1 P_3(x) P_3(x) dx &= \frac{2 \cdot 1}{2 \cdot 3 + 1} = \frac{2}{7} \\ \int_{-1}^1 P_3(x) \cdot P_4(x) dx &= 0 \\ \int_{-1}^1 (35x^4 - 30x^2 + 3)^2 dx \\ \Rightarrow P_4(x) &= \frac{1}{8} (35x^4 - 30x^2 + 3) \\ 64 \int_{-1}^1 [P_4(x)]^2 dx &= 64 \int_{-1}^1 P_4(x) \cdot P_4(x) dx = 64 \cdot \frac{2}{2 \times 4 + 1} = \frac{128}{9} \end{aligned}$$